

练习二十八 波动光学（二）

1. 2.5λ , 5

2. 625nm

3. (3); 4. (4)

5. $d \sin \theta = k\lambda$; $d = \frac{10^6}{500} = 2000(nm)$;

$$\Delta\theta = \arcsin \frac{2\lambda_1}{d} - \arcsin \frac{2\lambda_2}{d} = \arcsin 0.5896 - \arcsin 0.589$$

6. $I_1 = \frac{1}{2} I_0 \cos^2 60^\circ$; $I_2 = \frac{1}{2} I_0 \cos^2 30^\circ \cdot \cos^2 30^\circ$; $I_2 = \frac{9}{4} I_1$